General Mathematics — Exercise Sheet 4

Order and completeness in sequence limits

Variant: Statements

Exercise 1 (A recursive sequence). A real sequence satisfies $a_1 = 1$ and $a_{n+1} = \sqrt{2 + a_n}$ for $n \in \mathbb{N}$. Prove that it is well defined, nondecreasing and bounded above by 2. Then prove convergence and determine its limit. Use monotone convergence and the limit laws from Lectures 3–4; do not assume the limit exists when solving an equation for it.

Exercise 2 (A rational sequence with an irrational limit). A sequence satisfies $a_1 = 1$ and

$$a_{n+1} = \frac{1}{2}\left(a_n + \frac{2}{a_n}\right) \qquad (n \in \mathbb{N}).$$

Prove that every term is a positive rational number. Prove that the sequence from its second term onward is nonincreasing and bounded below by $\sqrt{2}$. Determine its real limit and explain why it has no rational limit. Use Lecture 1's irrationality of $\sqrt{2}$ and Lecture 3's uniqueness of real limits.

Exercise 3 (The Cauchy criterion). A real sequence (a_n) is called *Cauchy* if

$$\forall \varepsilon > 0 \quad \exists N \in \mathbb{N} \quad \forall m, n \geq N : \quad |a_m - a_n| < \varepsilon.$$

Prove that a real sequence converges if and only if it is Cauchy. For the reverse direction, first prove boundedness and then use Bolzano–Weierstrass from Lecture 4. Lecture 2 supplies the triangle inequality; Lecture 3 defines convergence.